

Windows 11 RTKLIB NTRIP Caster with Tailscale

Suggested workflow for delivering RTCM GNSS corrections from a Windows 11 base station using RTKLIB (Demo5 / rtkexplorer) as an NTRIP caster and Tailscale as the VPN.

1. Overview

Use **rtklibexplorer / Demo5** RTKLIB (not stock RTKLIB). Only that build's str2str / strsvr can act as an NTRIP **caster** (ntripc://). Tailscale gives rovers a stable VPN IP without router port-forwarding or a public IP.

2. Base station → Windows

1. Mount the base with a known ECEF/LLH position (surveyed, not autonomous).
2. Output **RTCM3** on serial/USB. Prefer RTCM-only (avoid mixing UBX/NMEA if possible).
3. Typical messages: **1005/1006** (base position) + MSM or legacy observations (**1074/1084/1094/1124** or **1004/1012**), plus **1033** if you want antenna/receiver info.
4. Note the COM port and baud rate (for example COM3, 115200).

3. Install RTKLIB (Demo5) and Tailscale

1. Download Windows binaries from <https://github.com/rtklibexplorer/RTKLIB/releases>.
2. Install Tailscale on the base PC and on every rover machine, phone, or field tablet that will pull corrections (<https://tailscale.com>).
3. Log all devices into the **same Tailnet**. On the base PC, note its Tailscale IP (100.x.y.z) from the Tailscale tray icon or `tailscale ip -4`.

4. Run the NTRIP caster

CLI (str2str) — from PowerShell in the RTKLIB bin folder:

```
str2str -in serial://COM3:115200:8:n:1:off -out ntripc://:2101/BASE
```

With basic authentication (recommended on a shared Tailnet):

```
str2str -in serial://COM3:115200:8:n:1:off -out ntripc://user:pass@:2101/BASE
```

Meaning:

- Input: base RTCM on COM3
- Output: local NTRIP caster on TCP **2101**, mountpoint **BASE**
- Empty host before :2101 means listen on all interfaces (including Tailscale)

GUI alternative: open strsvr, set Input = Serial (COM + baud), Output = **NTRIP Caster**, port 2101, mountpoint BASE, optional user/password, then Start.

Keep that window or process running while you need corrections.

5. Windows Firewall

1. Windows Security → Firewall → Advanced settings → Inbound Rule → Port → TCP 2101
2. Prefer scoping the rule to the Tailscale adapter or `100.64.0.0/10` so the caster is not exposed on LAN/Wi-Fi.
3. Confirm Tailscale is Connected and that Tailnet ACLs do not block inbound connections to the base PC (default Tailnets usually allow device-to-device).

6. Rover / client connection

On any Tailscale-connected device, point the NTRIP client at:

Field	Value
Caster host	Base Tailscale IP, e.g. <code>100.x.y.z</code>
Port	2101
Mountpoint	BASE (case-sensitive)
User / password	Same as caster, or dummy values if auth is disabled
Format	RTCM3

RTKLIB client example:

```
str2str -in ntrip://user:pass@100.x.y.z:2101/BASE -out tcpsvr://:9000
```

Or configure `rtknavi` / the receiver's built-in NTRIP client with host = Tailscale IP and mount = BASE.

7. Quick checks

1. On the base PC: `netstat -an | findstr 2101` → should show LISTENING
2. From a rover on Tailscale: fetch the source table against `100.x.y.z:2101` and confirm mountpoint BASE
3. Rover should receive RTCM. Fix quality depends on baseline, base coordinates, and message set — not on Tailscale itself.

8. Practical notes

- **Why Tailscale:** rovers reach `100.x.y.z:2101` with no router port-forward, CGNAT, or public IP. Prefer Tailscale-only exposure when possible.
- **Auth:** use `user:pass` on `ntrip://`; Tailscale ACLs can further restrict which devices may talk to the base.
- **Persistence:** put the `str2str` command in a `.bat` file and run it at login, or schedule it with Task Scheduler so it restarts on failure.
- **Multiple rovers:** the Demo5 caster can serve several clients from one serial/TCP input. Do **not** fan out a VRS stream this way.
- **Base must stay fixed:** wrong 1005/1006 position produces wrong RTK solutions even if the caster appears to work.

Requirement: Use the `rtkexplorer` / Demo5 build of RTKLIB. Other RTKLIB packages often include NTRIP client/server push only, not a local caster (`ntrip://`).